Quiz 1 — Sample 2

DATA 412/612

Weeks 1–4: R and the tidyverse, pipes and functions, ggplot2, and dplyr rows, columns, and groups.

Write your answers in an R script. You may use R and RStudio. You may use your notes and the course website (`https://amousavi1.github.io/612/data-612.html`). You may not use Google, any other website, or an LLM. Load the tidyverse. `mpg` is in `ggplot2`.

1. Build this data frame and, with tidyverse verbs, report the type of each column and how many `NA`s each column contains.

```
scores <- tibble(
  name  = c("Ada", "Lin", "Grace", "Alan"),
  lab   = c("8", "10", "NA", "7"),
  score = c(9.5, NA, 8, 10)
)
```

2. Write a function `late_flag` that takes **one** number. Return `"on time"` when the number is less than or equal to 0, `"late"` when it is greater than 0 and at most 30, `"very late"` when it is greater than 30, and `NA` when the input is `NA`. Try it on `-5`, `10`, `45`, and `NA`.

3. Using `mpg`, keep only the classes `compact` and `midsize`. Make a boxplot of highway mileage (`hwy`) by drive (`drv`). Label the axes in words and use `theme_bw()`.

4. Add a column `total` equal to `cty + hwy`. For each manufacturer, report the number of cars and the mean of `total`. Keep only manufacturers with at least 5 cars, and sort by that mean, highest first.